

Homework 3 - due Friday, February 13

1. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions of sets.
 - (a) Prove that if $g \circ f$ is injective then f is injective.
 - (b) Prove that if $g \circ f$ is surjective then g is surjective.
 - (c) Find an example of $f : A \rightarrow B$ and $g : B \rightarrow A$ such that $f \circ g = id_B$ but neither f nor g is bijective.
2. Let R be a commutative ring, and $S \subseteq R$ be a multiplicative set.
 - (a) Review the definition of $S^{-1}R$, and prove that $+$ and $\cdot$ make $S^{-1}R$ into a commutative ring with identity $1_R/1_R$.
 - (b) If $0 \in S$, show that $S^{-1}R$ is the zero ring.
3. Let R be a commutative ring. Recall that we say R is **Noetherian** if every ideal is finitely generated. Prove that R is Noetherian if and only if every chain of ideals eventually stabilizes, i.e., if

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

is a chain of ideals, then $I_N = I_{N+1} = I_{N+k}$ for all $k > 0$ and some $N \in \mathbb{N}$.

4. Let R be a commutative ring, and let I_j be ideals of R , and let P be a prime ideal of R .
 - (a) Prove that if $I_1 \cdots I_r \subseteq P$, then $I_j \subseteq P$ for some j . (Hint: Use contradiction).
 - (b) By (4a), if $P \supseteq \bigcap_{j=1}^r I_j$, then P contains one of the ideals I_j . Prove or disprove: if $P \supseteq \bigcap_{j=1}^{\infty} I_j$, then P contains one of the I_j .
5. Let R be a commutative ring, and let N be the set of nilpotents of R , i.e.

$$N =: \{r \in R \mid r^n = 0 \text{ for some } n \in \mathbb{N}\}$$

This is often called the **nilradical** of R , and you proved in the last homework that it is an ideal of R . Prove that N is contained in every prime ideal of R .

6. A ring is called **local** if it has a unique maximal ideal. Let R be a commutative local ring. Prove that if $x \in R$, then either x or $1 - x$ is a unit.
7. Let R be a commutative ring, and let $I \subseteq R$ be a proper ideal. Prove that the set of prime ideals containing I has minimal elements. These are the **minimal primes** of I . (Hint: Use Zorn's Lemma.)
8. Let F be a field. Prove that F contains a unique smallest subfield F_0 and that F_0 is isomorphic to either $\mathbb{Q}$ or $\mathbb{Z}/p\mathbb{Z}$ for some prime p . F_0 is called the **prime subfield** of F . (Hint: Intersect all subfields of F).
9. Let P be a prime ideal of a commutative ring R , and let R_P be the localization of R at P .
 - (a) Prove that if $f \notin P$, then $f/1$ is a unit in R_P .
 - (b) Let $I \subseteq P$ be an ideal of R . Prove that $IR_P = \{a/b \in R_P \mid a \in I\}$ is a proper ideal of R_P .
 - (c) Prove that if I is an ideal of R and $I \not\subseteq P$, then $IR_P = R_P$.
 - (d) Prove that R_P is a local ring, and describe its maximal ideal.